Compile Result

```
===== HOSPITAL PATIENT MANAGEMENT SYSTEM ====
=
1. Admit patient
2. Discharge patient
3. Search patient
4. Display forward
5. Display backward
6. Bed status
7. Undo discharge (re-admit oldest discharged
)
8. Discharge history
0. Exit
Enter choice: 1
Enter patient ID: 10
Enter name: Krushna
Enter age: 15
Enter disease: dengue
Patient admitted. Bed number 1 allotted.

===== HOSPITAL PATIENT MANAGEMENT SYSTEM ====
=
1. Admit patient
2. Discharge patient
3. Search patient
4. Display forward
5. Display backward
6. Bed status
7. Undo discharge (re-admit oldest discharged
)
8. Discharge history
0. Exit
Enter choice: 1
Enter patient ID: 11
Enter name: parth
Enter age: 18
Enter disease: 12 difficency
Patient admitted. Bed number 2 allotted.

===== HOSPITAL PATIENT MANAGEMENT SYSTEM ====
=
1. Admit patient
2. Discharge patient
3. Search patient
4. Display forward
```

Compile Result

```
===== HOSPITAL PATIENT MANAGEMENT SYSTEM ====
=
1. Admit patient
2. Discharge patient
3. Search patient
4. Display forward
5. Display backward
6. Bed status
7. Undo discharge (re-admit oldest discharged
)
8. Discharge history
0. Exit
Enter choice: 4
Patients (forward, head to tail):
  ID: 10 | Name: Krushna | Age: 15 | Disease:
 dengue | Bed: 1
  ID: 11 | Name: parth | Age: 18 | Disease: 1
2 difficency | Bed: 2
Total patients: 2

===== HOSPITAL PATIENT MANAGEMENT SYSTEM ====
=
1. Admit patient
2. Discharge patient
3. Search patient
4. Display forward
5. Display backward
6. Bed status
7. Undo discharge (re-admit oldest discharged
)
8. Discharge history
0. Exit
Enter choice: 2
Enter patient ID to discharge: 10
Krushna discharged. Bed 1 is now free.

===== HOSPITAL PATIENT MANAGEMENT SYSTEM ====
=
1. Admit patient
2. Discharge patient
3. Search patient
4. Display forward
5. Display backward
6. Bed status
7. Undo discharge (re-admit oldest discharged
```

Compile Result

```
Enter choice: 1
Enter patient ID: 19
Enter name: Omkar
Enter age: 17
Enter disease: cough
Patient admitted. Bed number 1 allotted.

===== HOSPITAL PATIENT MANAGEMENT SYSTEM ====
=
1. Admit patient
2. Discharge patient
3. Search patient
4. Display forward
5. Display backward
6. Bed status
7. Undo discharge (re-admit oldest discharged
)
8. Discharge history
0. Exit
Enter choice: 6
  Bed 1: Occupied
  Bed 2: Occupied
  Bed 3: Free
  Bed 4: Free
  Bed 5: Free
  Bed 6: Free
  Bed 7: Free
  Bed 8: Free
  Bed 9: Free
  Bed 10: Free
Free beds: 8 / 10

===== HOSPITAL PATIENT MANAGEMENT SYSTEM ====
=
1. Admit patient
2. Discharge patient
3. Search patient
4. Display forward
5. Display backward
6. Bed status
7. Undo discharge (re-admit oldest discharged
)
8. Discharge history
0. Exit
Enter choice:
```